1. FinTech

1. NFC-Based Offline UPI Payments for Bharat 2.0

Problem Statement:
India’s digital payment revolution through UPI has largely benefited urban and semi-urban populations. However, in low-connectivity regions such as tribal belts, rural towns, or disaster zones, poor internet connectivity and low digital literacy limit its usability.

Challenge:
Build a NFC-enabled, offline-first UPI payment system for feature-rich Android devices that can store transaction intents locally and broadcast them securely via near-field communication. Once a connection is re-established, the queued transactions should sync and settle with banks.

Considerations:
• Tokenization of payments for offline security
• Zero connectivity reconciliation model
• Regulatory compliance with NPCI

2. Intelligent Portfolio Manager with Real-Time Risk Profiling

Problem Statement:
Retail investors often make ad-hoc investment decisions based on internet trends, social media, or influencers. This leads to unbalanced portfolios and poor returns in volatile markets.

Challenge:
Design an AI-driven Portfolio Management Tool that monitors live financial data, news sentiment, and user-defined constraints to suggest automated rebalancing strategies using risk-reward optimization.

Features:
• Portfolio risk prediction using VaR, CVaR
• Sentiment-driven rebalancing signals
• Explainable AI models for transparency

3. High-Frequency Market Making Bot (Jane Street Inspired)

Problem Statement:
Market-making firms like Jane Street use ultra-low-latency systems to provide liquidity across exchanges. Achieving competitive edge in such systems involves fine-tuned predictive models and millisecond-level execution infrastructure.

Challenge:

Develop a high-frequency trading (HFT) simulator and market-making algorithm that operates on millisecond price feeds, performs order book analytics, and quotes bids/asks dynamically for profit, while minimizing inventory risk.

Key Components:
• Real-time L2 order book analysis
• Volatility clustering for position sizing
• Latency optimization (threaded async engine)
• Dynamic spread quoting and adverse selection detection

Constraints:
• Backtest against historical NSE/BSE data
• Benchmark PnL, Sharpe ratio, and max drawdown
• Optional FPGA or GPU acceleration simulation

2. Ed-Tech

1. Immersive STEM Education Platform using AR/VR

Problem Statement:
Complex scientific phenomena like DNA replication, electromagnetism, or orbital mechanics are hard to teach using static visuals. Many students, especially in Tier-2/3 schools, lack exposure to experimental learning.

Challenge:
Create an AR/VR-based interactive learning platform that simulates textbook content into immersive 3D learning environments. Students should be able to interact with the environment using mobile phones or headsets and perform experiments virtually.

Key Ideas:
• Physics lab simulations (e.g., Ohm's Law circuits)
• Gesture or speech-based controls
• Teacher dashboard with assessment tracking

2. AI-Driven Subjective Answer Evaluation System

Problem Statement:
Evaluating essays and descriptive answers is time-consuming and subjective, especially when grading at scale (e.g., university exams, competitive tests).

Challenge:
Build a system that uses large language models to automatically evaluate and score long-form answers based on contextual correctness, coherence, grammar, and depth.

Core Tasks:
• Compare student answers to gold-standard key points
• Grade with explainability (rubrics, feedback)
• Detect off-topic or hallucinated answers

3. Research Paper Conference Management ToolKit (CMT) for AI-Generated and Plagiarized Content

Problem Statement:
With the rise of AI writing tools (e.g., ChatGPT), plagiarism in school and college submissions has evolved. Standard tools like Turnitin fail to detect AI-generated original content. Thus develop a **Conference Management ToolKit** for research paper conference

Challenge:
Develop a CMT suite that flags submissions for textual plagiarism, stylistic inconsistency, and AI-generated language patterns.

Core Modules:
• Stylometric fingerprinting per student
• AI-watermark detection using entropy and perplexity
• Plagiarism across multiple languages and rephrased content

3. MedTech

1. Virtual Doctor Assistant for Remote Areas

Problem Statement:
Rural populations often delay primary care due to inaccessibility of doctors or fear of high costs, leading to preventable complications.

Challenge:
Design a virtual healthcare assistant that can collect basic symptoms, patient history, and lifestyle details through a chatbot interface and suggest initial diagnosis, treatment paths, and connect with telemedicine portals.

Bonus: Multilingual and audio support for non-literate users using Whisper

AI Stack:
• NLP for symptom extraction
• Medical triage decision trees + GPT-4 for contextual responses
• Integration with India’s Ayushman Bharat Digital Health Mission

2. Schizophrenia Detection using EEG and Deep Learning

Problem Statement:
Schizophrenia diagnosis is delayed due to reliance on behavioral cues. EEG-based bio-markers can help detect it earlier, but models are underdeveloped.

Challenge:
Create a deep learning pipeline that processes EEG signals to identify neurophysiological patterns linked to early schizophrenia.

Pipeline Requirements:
• Raw EEG signal cleaning
• Feature extraction using Wavelet + Fourier
• CNN/Transformer-based classification
• Training on open EEG datasets (e.g., TUH EEG Corpus)

3. Mental Health Companion for Gen-Z

Problem Statement:
Post-pandemic anxiety, depression, and burnout have spiked among students and professionals, but stigma prevents them from seeking help.

Challenge:
Build an AI-powered mental wellness chatbot trained on CBT (Cognitive Behavioral Therapy) and mindfulness practices that offers conversational support, journaling prompts, and emotional tracking.

Functions:
• Mood journaling and daily mental check-ins
• Emotion detection from text or voice
• Suggest personalized coping strategies or meditations

4. SDG (Sustainable Development Goals)

1. Early Flood Warning and Detection using LiDAR

Problem Statement:
Floods cause devastating damage to human life, agriculture, and infrastructure, especially in rural and coastal areas where early detection systems are limited.

Challenge:
Develop a LiDAR-powered flood early-warning system that maps terrain, predicts water flow accumulation, and issues timely alerts to reduce damage and improve disaster response.

Unique Features:
• Real-time LiDAR data integration with GIS
• Terrain-based flood risk modeling
• Mobile-based community alert system

2. Pollution Detection using GIS + Satellite Imagery

Problem Statement:
Monitoring air and water pollution at scale requires costly ground sensors. Satellite and GIS data provide untapped potential for large-scale analysis.

Challenge:
Build a system that uses Sentinel-5P, MODIS, and GIS layers to predict pollution levels (PM2.5, NO2) across regions and forecast risk zones.

Bonus:
• Crop burning impact modeling
• Temporal anomaly detection
• Visual GIS dashboard for public policy makers

3. CME Detection and Alert System using Aditya-L1 Data (you may use other data also)

Problem Statement:
Coronal Mass Ejections (CMEs) can disrupt communication satellites, navigation systems, and even power grids. Aditya-L1 provides invaluable solar data.

Challenge:
Design a tool that processes Aditya-L1 ASPEX (Solar wind data) and SUIT payloads to detect CME events, classify their intensity, and forecast their potential Earth impact.

Key Deliverables:
• Time-series anomaly detection using solar wind speed, particle flux
• CME direction and strength estimation
• Alert system for satellite operators or ISRO ground stations
• Optional: Correlation with CACTus database and space weather APIs